

ZenTask

Introduction

The individual project titled “ZenTask – Smart Task Search & Filter” is an enhancement of the existing ZenTask system — a collaborative project management platform designed to help teams efficiently create, assign, and monitor their tasks.

While ZenTask already provided effective task management, team collaboration, and analytics, it lacked an intelligent and dynamic way for users to quickly locate specific tasks or filter information within busy project boards.

To address this limitation, the Search and Filter feature was developed.

This feature allows users to instantly search, categorize, and narrow down tasks based on task names, assigned members, or progress — making ZenTask more user-friendly, data-efficient, and productivity-oriented.

Identifying Business Problem from the Group Project

In the group project ZenTask, users could successfully create boards, assign tasks, and track project progress.

However, as the number of tasks grew, it became increasingly difficult for team members to locate or manage specific tasks quickly.

This limitation created several real-world challenges:

- Users had to scroll through long task lists to find a particular task.
- Filtering tasks by assigned member or completion status was not possible.
- Managers struggled to track progress efficiently in large projects.
- Time was wasted navigating through cluttered task boards.

These usability issues affected workflow speed, clarity, and overall team productivity.

Objective of the New Feature

The Smart Search and Filter feature was developed with the following objectives:

1. Enable Quick Task Discovery: Allow users to search tasks instantly by name or keyword.
2. Add Dynamic Filtering Options: Provide dropdown filters to view tasks assigned to specific members.
3. Enhance Data Navigation: Simplify how users browse through ongoing, completed, or assigned tasks.
4. Improve User Efficiency: Reduce manual scrolling and make task retrieval effortless.
5. Integrate Seamlessly: Ensure smooth integration with the existing dashboard without performance loss.

Requirements

Functional Requirements

- Provide a responsive search bar to find tasks by name or keyword.
- Add a dropdown menu to filter tasks based on assigned team members.
- Dynamically show or hide tasks based on search and filter input without page reload.
- Ensure instant updates on the frontend using JavaScript event listeners.
- Maintain full compatibility with Flask backend and existing HTML templates.

Non-Functional Requirements

- Responsive design compatible with desktop and mobile.
- Minimal latency for real-time filtering.
- Smooth animations and transitions for better UX.
- Efficient DOM rendering to handle large task datasets.
- Secure and optimized for all user roles (admin, manager, or member).

Tools & Technologies

- Frontend: HTML5, CSS3, JavaScript (DOM manipulation)
- Backend: Flask (Python)
- Database: PostgreSQL
- Libraries: Jinja2 for rendering dynamic task data

Technical Work and Analysis

The Search and Filter feature was implemented by enhancing the frontend task rendering logic within the dashboard and board views of ZenTask.

Technical workflow:

1. The search input and member filter dropdown are placed above the task grid.
2. JavaScript event listeners detect user typing or filter selection.
3. The script compares each task card's text (name, assigned member) against the search and filter criteria.
4. Tasks that don't match the conditions are dynamically hidden, while matching ones remain visible.
5. All operations happen instantly in the browser — no page reload or database query required.

Integration highlights:

- Compatible with Flask's Jinja2 templating (auto-populates member list).
- Maintains the current board structure and drag-drop features.
- Optimized for fast filtering even on boards with 100+ tasks.
- Designed with accessible UI and intuitive controls.

Solving the Business Problem

The Search and Filter update directly addresses the inefficiency of navigating through long task lists.

With this feature:

- Users can instantly locate tasks by typing part of a task name.
- Team leaders can view only those tasks assigned to a specific member.
- Completed or important tasks can be found without manually scanning the entire board.
- The feature improves data accessibility and reduces task management time significantly.

By simplifying navigation, the Search and Filter feature enhances team productivity, focus, and clarity — turning ZenTask into a truly efficient task management platform.

Result of the New Feature Update

After adding the Smart Search and Filter system:

- Users can now search tasks instantly by keyword.
- Filtering by member or criteria happens smoothly without reloading.
- Time spent locating tasks decreased drastically.
- The overall UI feels faster, cleaner, and more interactive.
- Managers can track specific team member progress in seconds.

This feature makes ZenTask more organized, responsive, and user-focused.

Conclusion

The ZenTask – Smart Task Search & Filter feature effectively transforms the task management experience by introducing intelligent navigation and instant data retrieval.

By combining HTML, CSS, JavaScript, and Flask, the feature enables real-time, efficient filtering without backend delays.

It resolves a key usability gap in the original project by improving accessibility and productivity for all users.